Neurology and autonomic regulation

Whole system foundations and evidence limits

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The nervous system links perception, action and internal regulation through interacting circuits. This reference explains that larger architecture, then examines sympathetic, parasympathetic and enteric organization in detail. Its central conclusion is that autonomic physiology is organ-specific and context-dependent, rather than a universal activation-relaxation switch. [1]

The report is intended for careful study and reference, including scrutiny of claims that borrow neurological language. It does not provide a diagnosis, treatment plan or self-testing protocol. Research interpretations and proposed uses in Blackwish are kept separate. No project revision is implemented or ratified by this document.

Reading guide

- The two-page executive summary identifies the main distinctions and cautions.
- Neural foundations explain the map from cells and circuits to sensing, movement, learning and sleep.
- Autonomic organization compares pathways, ganglia, transmitters and receptors.
- Organ regulation shows why apparently simple rules have important exceptions.
- Clinical measurement distinguishes an observation, a syndrome, a cause and a human interpretation.
- The final sections provide a glossary, a research roadmap and three textual consistency checks.

Evidence is identified as established anatomy, synthesis, a specific human or animal study, a clinical guideline, or a contested interpretation where that distinction matters. The reference list links to the underlying sources. Older foundational work is used for stable concepts; newer findings do not automatically replace established clinical terminology.

Executive summary

A useful map of the nervous system

The central nervous system comprises the brain and spinal cord. Peripheral nerves and ganglia connect these structures with the body. This anatomical map differs from the functional map of sensory input, motor output and internal regulation. Afferent and efferent describe signal direction, not conscious versus unconscious activity. [2]

Complex behavior depends on interacting regions and circuits. Cortex, basal ganglia and cerebellum work within connected systems rather than isolated boxes for personality traits. Mechanisms at one level do not automatically explain observations at another: a receptor response, a network pattern and a person's experience are different kinds of evidence. [3]

Neural change is possible, but plasticity does not promise unlimited improvement. Learning involves several mechanisms and timescales. Memory, executive function and sleep each require their own distinctions; they cannot be collapsed into one measure of capacity or willpower. [4], [5]

The autonomic distinction that matters most

Sympathetic and parasympathetic labels chiefly classify visceral motor output. They do not classify every sensory fiber traveling alongside that output. The vagus is a mixed nerve with important sensory as well as parasympathetic components; the enteric nervous system also performs local processing in the gut. [6]

The usual autonomic output route has a preganglionic neuron, a peripheral ganglion and a postganglionic neuron. Somatic motor output to skeletal muscle uses a different arrangement. Automatic breathing and spinal reflexes therefore cannot be labeled parasympathetic merely because they proceed without deliberation. [7]

Both autonomic divisions use acetylcholine in their preganglionic pathways. Most sympathetic postganglionic signaling uses norepinephrine; parasympathetic postganglionic signaling commonly uses acetylcholine. Receptors and target tissues determine the response. Eccrine sweat glands are a prominent sympathetic cholinergic exception. [8], [9]

Executive summary continued

Organ specific exceptions are part of the rule

Example	What it prevents us from assuming
Heart and vessels	Slower pulse is not a complete measure of cardiac function, and vascular dilation need not require parasympathetic innervation.
Human airways	Circulating catecholamines and direct sympathetic innervation are different routes.
Gut and bladder	Successful function may require contraction in one tissue and relaxation in another. Bladder parasympathetic output is pelvic, not vagal.
Skin	Sweating is sympathetic cholinergic; temperature regulation also involves specialized vascular responses.

These examples are developed in the organ chapters. They show why a claim should name its organ, pathway, receptor, timescale and measured outcome. [10], [11], [12], [13]

Measurements do not supply every interpretation

Brain imaging, electrical recordings, peripheral nerve studies and autonomic tests examine different properties. A normal result in one domain cannot exclude every neurological explanation. Autonomic evaluation itself separates cardiovascular adrenergic, cardiovagal and sudomotor domains; no single test establishes the entire distribution of autonomic failure. [14]

Heart-rate variability and respiratory sinus arrhythmia can be useful cardiac measurements, but neither is a recording of all vagal output. In a primary study, breathing materially affected the relationship between RSA and pharmacologically assessed cardiac parasympathetic control. A physiological proxy alone cannot establish trust, consent, moral character or the meaning of an experience. [15]

Keep contested mechanisms distinct from established anatomy

Developmental studies in mice challenge conventional sacral classification, but the proposed reclassification remains disputed. The eLife assessment of a 2024 study judged evidence for its modified-sympathetic interpretation incomplete. This reference retains conventional human clinical terminology. [16]

Polyvagal theory also contains contested physiological and evolutionary propositions. A theory's explanatory claims, an intervention's outcome evidence and a person's account of their experience must be evaluated separately. Disagreement about one does not settle the others. The measurement chapter presents the critique and response without treating their existence as equal evidential support. [17], [18]

Neural foundations

The nervous system is not a single command center with a few chemical mood switches. It is a living, energy-dependent collection of interacting control systems. A useful reading order is **anatomy to cells to circuits to functions to behavior**, while remembering that explanations at one level do not automatically establish explanations at another. A receptor mechanism does not by itself explain a personal decision. This chapter supplies the map; sympathetic and parasympathetic organization is developed next.

Two maps that should not be confused

Central versus peripheral is primarily an anatomical distinction. The central nervous system (CNS) comprises brain and spinal cord. The peripheral nervous system (PNS) connects central structures with receptors, muscles and organs through nerves and ganglia. A neuron is a cell; a nerve is a bundle of peripheral axons, not one enormous neuron. Afferent signals travel toward the CNS; efferent signals travel outward. These directions are relative to the central system, not synonyms for conscious and unconscious. [2]

Sensory, motor and regulatory are functional distinctions. Somatic motor pathways control skeletal muscle; autonomic pathways participate in visceral regulation. Neither category operates independently of central processing. Sensory feedback continually changes motor and regulatory output. A reflex can organize action without waiting for conscious deliberation: the spinal cord contains processing circuits as well as ascending and descending pathways. Calling it merely a cable misses an important part of the system. [19]

There are conventionally twelve pairs of cranial nerves, carrying different combinations of sensory, motor and autonomic functions. Most have major connections with the brainstem; the optic pathway has CNS characteristics, and the spinal accessory nerve has a spinal contribution. Cranial nerves therefore resist a perfectly tidy "everything outside the brain is PNS" diagram. Their combined functions include vision, hearing, balance, facial sensation and movement, swallowing, and communication with internal organs. [20], [21]

Cells and communication

Neurons communicate through specialized synapses. Many conduct action potentials along axons. Chemical synapses release transmitters through calcium-dependent processes; electrical synapses directly couple cells. Their properties shape signaling. [22]

Glia are active participants, not packaging. Astrocytes help regulate extracellular ions, transmitter clearance and metabolic support. Oligodendrocytes form CNS myelin, changing how reliably and rapidly axons conduct signals. Microglia participate in immune surveillance and remodeling of connections. Their roles vary with development, region and physiological conditions. Evidence that glia influence learning does **not** imply that any particular glial mechanism explains a person's experience; much mechanistic evidence comes from animals or cellular preparations. [23]

Excitation and **inhibition** concern a postsynaptic cell's likelihood of firing, not excitement and calm. Glutamate and GABA commonly support excitation and inhibition respectively in mature CNS circuits; receptor, ionic and cellular context matter. [24]

At the circuit level, inhibiting an inhibitory neuron can release another population from suppression: **disinhibition**. For an illustrative circuit, if A inhibits B and B inhibits C, increasing A can increase C's

output. This is a logical example, not a claim about a specific behavior. It shows why the effect of a molecule cannot be read straight off its name. Neuromodulatory systems additionally alter circuit responsiveness, learning and state over different spatial and temporal scales; dopamine, serotonin, acetylcholine and noradrenaline do not each correspond to one emotion. [25]

Dopamine is an especially important corrective example. In well-studied reward-learning tasks, phasic activity in many dopamine neurons reflects differences between obtained and predicted outcomes. Such signals can help update future expectations. That is not identical to pleasure, and it does not describe every dopamine neuron or every dopamine function. Distinguish the stimulus, its learned predictive value, the teaching signal, the resulting action, and the subjective experience. Collapsing them into "a dopamine hit" discards the mechanism being explained. [26]

Brain regions are nodes not personality boxes

The cerebral cortex contains specialized but interacting territories. Occipital regions are central to visual processing; temporal regions contribute to hearing and other functions; parietal regions integrate sensory and spatial information; frontal regions contribute to planning, action and complex cognition. Brainstem circuitry supports basic life functions, and cerebellar circuitry contributes to coordination. These are useful orientation points, not exclusive ownership claims: a complex task recruits several systems. [27]

Deeper structures extend this map. The thalamus participates in organized communication between brain systems. The hypothalamus links neural regulation with bodily needs and endocrine signaling. Hippocampal circuitry is important for forming and organizing particular kinds of memory. The basal ganglia and cerebellum interact with cortex and with each other; their roles extend beyond movement into learning and cognition. The useful question is usually "which interacting circuit, performing which computation?" rather than "where does this trait live?" [28], [3]

Sensation action and bodily regulation form loops

Sensation is not limited to the familiar five senses. Proprioceptive signals concern body configuration and movement; vestibular signals concern motion and orientation. Interoception concerns the body's internal state and the processes that sense, interpret and regulate it. Neural routes interact with hormonal, immune and other non-neural communication. Many internal signals never become conscious sensations. Conversely, a consciously felt bodily change does not uniquely identify its physiological cause. [29]

Motor behavior likewise combines central plans, sensory feedback and ongoing adjustment. Consider reaching for a cup: the target must be located, posture coordinated, an action selected, muscle activity organized, and movement corrected when the cup is heavier than expected. This is a deliberately simplified illustration of distributed control, not a fixed sequence of isolated modules. Basal-ganglia and cerebellar loops contribute different but interacting operations; neither is merely an optional accessory to the cortex. [30]

Plasticity learning and memory

Plasticity means that nervous-system properties can change; it does not mean unlimited malleability or guaranteed improvement. Changes can occur in synaptic efficacy, structural connectivity and myelination, with different mechanisms and timescales. A nervous system must remain sufficiently stable to preserve useful skills while sufficiently adaptable to learn. Recent neuron-glia research

broadens the cellular account of this adaptation, but many causal details and their relevance to human behavior remain unresolved. [4]

Memory is also plural. Keeping information available briefly differs from establishing durable knowledge; remembering an episode differs from acquiring a motor skill. For many declarative memories, hippocampal and cortical processes interact during encoding and subsequent consolidation. Consolidation includes cellular stabilization and longer-term reorganization across systems. It is not literally a file being moved intact from one brain compartment to another. Different theories disagree about the continuing hippocampal contribution to detailed remote memories, so "all old memories leave the hippocampus" is too strong. [31]

Sleep and executive function

Sleep is an actively regulated state, not simply the brain switching off. Sleep pressure accumulated during wakefulness interacts with circadian timing, which is influenced by environmental cues including light. REM and non-REM sleep differ in their physiology, and their organization changes across the night and lifespan. These distinctions matter because duration, timing and sleep architecture are related but not interchangeable variables. [32], [33]

Executive function describes interrelated capacities including working memory, inhibitory control and cognitive flexibility. These support holding a goal in mind, resisting a competing response and changing strategy when conditions change. Their performance is sensitive to task demands and physiological or emotional context. They are not a single reserve of willpower, and one difficult afternoon does not diagnose a stable deficit or a person's character. Sleep is relevant because inadequate sleep can affect learning, thinking and response performance. [5], [34]

Constraints and interpretive discipline

Neural computation has a material cost: maintaining gradients, transmitting signals and supporting cellular maintenance require energy. Energy-budget estimates depend on cell types and model assumptions; they should not be converted into precise personal "mental battery" percentages. Likewise, increased activity is neither inherently better nor automatically evidence of greater capacity. [35]

Finally, distinguish **observation, association, causal intervention and explanatory model**. A brain region associated with a task may participate in many other tasks. Its activation alone therefore cannot securely identify a particular thought, emotion or motive. Animal experiments can establish mechanisms under controlled conditions without proving a universal human explanation. The most defensible whole-system account keeps anatomy, measured physiology, behavior and lived experience connected - but does not pretend that one can be unambiguously translated into another. [36]

Autonomic organization

Comparison of the main output pathways

Feature	Sympathetic	Parasympathetic
Conventional central origin	Thoracic and upper lumbar cord	Brainstem and sacral cord
Typical peripheral relay	Sympathetic chain or prevertebral ganglia	Ganglia near or within target organs
Preganglionic transmitter	Acetylcholine	Acetylcholine
Common postganglionic transmitter	Norepinephrine	Acetylcholine
Receptors at typical targets	Adrenergic alpha and beta subtypes	Muscarinic subtypes
Important caution	Sweating is mainly sympathetic cholinergic	The vagus is not the entire division
Relation to the other division	May vary reciprocally or independently	May vary reciprocally or independently

This is a conventional organizational summary, not a rule that every tissue has opposing wires. The adrenal medulla is a neurosecretory exception, and organ chapters identify other departures. Transmitter and receptor details are discussed below. [8], [37]

What is being divided and what is not

Afferent signals travel toward the CNS; **efferent** commands travel away. Sensory axons accompanying autonomic output are not themselves sympathetic or parasympathetic motor neurons. The vagus is a mixed nerve, so describing the entire nerve as parasympathetic obscures its sensory components. [6]

The basic somatic motor route uses a lower motor neuron projecting from the central nervous system to skeletal muscle. The usual autonomic efferent route instead has a central **preganglionic neuron**, a peripheral ganglionic synapse, and a **postganglionic neuron** reaching an organ. This distinction concerns circuitry, not whether every action is consciously chosen: somatic spinal reflexes can occur automatically. A ganglion is a peripheral collection of neuronal cell bodies; a nerve is an axon bundle, and a plexus is an interwoven network. [7]

Anatomical comparison

Sympathetic outflow originates in thoracic and upper-lumbar spinal regions, approximately T1-L2/L3. Preganglionic axons reach sympathetic-chain or prevertebral ganglia. Ascending and descending distribution permits targets throughout the body. [38]

Parasympathetic outflow conventionally originates in brainstem and sacral regions. Cranial routes include III, VII, IX and X; sacral pelvic pathways supply pelvic viscera. Ganglia usually lie near or within targets. The vagus is only one component. [39]

The **adrenal medulla** is a crucial exception to the ordinary two-neuron diagram. Sympathetic preganglionic fibers activate chromaffin cells, which release catecholamines (epinephrine and norepinephrine) into blood rather than extending a conventional postganglionic axon to a single target. Neural sympathetic output and circulating adrenal hormones are therefore related but not identical mechanisms. Likewise, being innervated by a mixed peripheral nerve does not erase distinctions among sensory, somatic motor, and autonomic axons traveling inside it. Organ-level diagrams should name the relevant fiber population, not assign a single function to an entire nerve. [40]

Transmitters and receptors the exceptions matter

Both sympathetic and parasympathetic preganglionic neurons release **acetylcholine (ACh)**. Fast transmission in autonomic ganglia involves neuronal nicotinic receptors. Most sympathetic postganglionic neurons use **norepinephrine**, while conventional parasympathetic postganglionic transmission commonly uses ACh at muscarinic receptors. These labels describe transmitter-receptor relationships, not a universal rule that ACh inhibits and norepinephrine excites. Effects depend on receptor subtype, target cell, and downstream signaling. "Cholinergic" is therefore not synonymous with "parasympathetic." [8]

Nicotinic receptors are ligand-gated ion channels; muscarinic receptors are G-protein-coupled receptors. Neuronal nicotinic receptors in autonomic ganglia should be distinguished from muscle-type nicotinic receptors at skeletal neuromuscular junctions. Muscarinic receptor subtypes also differ across tissues: a transmitter's name alone is insufficient to predict an organ's response. A pathway description should specify both **where the signal originates** and **which receptor receives it**. That avoids the common error of treating a chemical as if it carried one fixed physiological meaning everywhere. [37]

Eccrine sweating is sympathetic but predominantly cholinergic: its postganglionic fibers release ACh acting on muscarinic receptors, rather than following the usual noradrenergic sympathetic pattern. Autonomic signaling additionally includes nitric oxide, ATP, and peptide cotransmitters in target-specific pathways. The useful organizational unit is often the projection to a particular tissue, not an undifferentiated division. Baseline (**tonic**) and event-linked (**phasic**) activity occur in both divisions; sympathetic pathways can be active at rest, and parasympathetic output can occur in discrete bursts. [1]

Adrenergic receptors likewise comprise multiple subtypes rather than a single "adrenaline receptor." Alpha and beta receptor families have different distributions and signaling properties; beta-1 receptors are prominent in cardiac tissue, while beta-2 receptors are important in the lung. Consequently, the same circulating catecholamine can have different effects in different tissues. Receptor biology is one reason that an increase in a broad transmitter measure cannot be translated directly into a single whole-body functional state. [41]

Central autonomic coordination

Sympathetic and parasympathetic outputs are coordinated through a **central autonomic network**, not two isolated command systems. The nucleus tractus solitarius receives visceral sensory input; medullary circuits help organize cardiovascular and respiratory regulation. Hypothalamic circuits connect bodily regulation with endocrine functions. Insular, anterior cingulate and amygdala networks participate in integrating bodily information with context and action. Spinal reflex circuits interact with descending control and ascending feedback through brainstem and forebrain pathways. These are reciprocal, state-dependent loops rather than a simple top-down hierarchy. Naming a region does not

identify a unique emotion or reveal someone's motive; each region contains diverse populations and participates in multiple functions. [42], [43]

Coordination not a binary switch

Reciprocity is only one possible pattern. In a human study of simulated diving, investigators studied 56 healthy volunteers and found increased muscle sympathetic nerve activity, blood pressure, and vascular resistance, with slower heart rate emerging in the later phase. The sympathetic signal was measured using microneurography; cardiac parasympathetic involvement was inferred from the physiological response rather than a direct vagal recording. This provides evidence against a universal either/or switch, but does not establish one identical response in every organ or participant. It is evidence about physiology, not an instruction to reproduce the protocol. [44]

Enteric networks are more than a parasympathetic relay

The **enteric nervous system (ENS)** contains gut-wall neurons and glia. Myenteric networks regulate motility; submucosal networks contribute to secretion and local blood flow. Sensory neurons, interneurons, and motor neurons support local processing, modified by extrinsic inputs. Many autonomic effects on intestinal motility and secretion operate through enteric circuits. Exceptions include direct sympathetic innervation of submucosal arteries and sphincteric smooth muscle. Thus, "parasympathetic activation increases digestion" is only a mnemonic, not a complete circuit description. [6]

Why sacral classification remains disputed

Conventional human teaching identifies S2-S4 pelvic outflow as parasympathetic. A 2016 mouse developmental study challenged that assignment: sacral pre- and postganglionic neurons shared a panel of developmental and phenotypic features with thoracolumbar sympathetic neurons rather than cranial parasympathetic populations. Its authors proposed a cranial-versus-spinal reclassification. These findings concern developmental identity and classification; they do not mean the previously mapped pelvic pathways disappeared or that their functional effects were newly discovered. [45]

Critics argue that developmental similarities do not, by themselves, supersede functional-anatomical criteria. In particular, sharing spinal developmental programs is not necessarily equivalent to belonging to one functional autonomic division. Horn's 2018 analysis defended retaining the parasympathetic designation and called for further experiments. The disagreement therefore involves which evidence should define a category, as well as observations about the neurons. A careful account reports both the conventional terminology and the challenge instead of presenting one provocative paper title as consensus. [46]

A 2024 mouse single-cell study extended the reclassification argument, finding pelvic ganglion classes distinct from typical sympathetic and cranial parasympathetic populations, with greater genetic kinship to sympathetic groups. However, the paper's own **eLife editorial assessment judged the evidence for the modified-sympathetic interpretation incomplete**. This is an especially useful warning against overclaiming: modern molecular methods can reveal important heterogeneity without automatically resolving nomenclature or establishing a universal human conclusion. This reference therefore retains conventional clinical labels and flags the alternative explicitly. [16]

Organ regulation

Coordinated organ specific outputs

The useful question is not simply "Which branch is on?" but **which pathway, acting on which tissue and receptor, under which physiological conditions?** Sympathetic and parasympathetic activity can change reciprocally, rise together, fall together, or vary relatively independently. These descriptions refer to particular outputs over particular intervals - not necessarily to the entire body. Human research using cardiac sympathetic and parasympathetic indices illustrates why a single activation-relaxation line is inadequate; importantly, such indices are indirect observations, not direct readings of every autonomic nerve. [47]

Heart vessels and blood pressure feedback

Heart: sympathetic norepinephrine acting principally through beta-1-adrenoceptors generally accelerates pacemaking and conduction, strengthens contraction and speeds relaxation. Parasympathetic acetylcholine acting principally through M2 receptors slows pacemaking and atrioventricular conduction. These are different controllable variables: a heart-rate value is not a complete description of cardiac function. Local cardiac ganglia participate in a larger network that also includes extracardiac ganglia, spinal cord and brain. Modern evidence also makes the old claim that the ventricles have essentially no parasympathetic innervation too absolute. [10]

Vessels: the main neural control of most vascular beds is sympathetic, often maintaining a background constrictor influence. Reducing this influence can permit dilation without recruiting a matching parasympathetic nerve. Actual diameter also reflects local tissue demand and endothelial mechanisms; "more sympathetic activity" does not imply identical flow changes everywhere. The distinction between neural input and the final vascular response is essential. [48]

Arterial baroreflex: pressure-related stretch in the carotid sinus and aortic arch changes sensory firing. Carotid signals reach the brainstem through cranial nerve IX; aortic signals travel through X. Brainstem processing can increase cardiac parasympathetic output while reducing sympathetic cardiac and vascular output when pressure rises. This buffers the disturbance through changes in cardiac performance and vascular resistance. A pressure fall generally evokes the converse pattern. The IX/X signals entering the brain are **afferent information**; the returning cardiac and vascular commands are **efferent output**. Sharing a named nerve does not make those functions identical. [49]

Feedback is context-sensitive rather than rigid. During exercise the baroreflex continues operating, but its operating range shifts toward the higher pressure appropriate to the activity. Human experiments show contributions from central motor command and muscle-derived feedback. Thus, a pressure increase during activity is not evidence that the baroreflex has simply switched off. [50]

Breathing is not the same as airway autonomic control

The respiratory pump and the bronchial tubes require separate explanations. Brainstem respiratory networks drive spinal motor neurons supplying breathing muscles; the diaphragm is activated through the phrenic motor system. Automatic operation does not turn a skeletal-muscle motor pathway into a parasympathetic pathway. Sensory and descending inputs continuously shape this motor output. [51]

Chemical feedback contributes information about oxygen, carbon dioxide and acidity. Peripheral chemoreceptors and central CO2/pH-sensitive systems influence respiratory networks, while

pulmonary sensory input adds mechanical information. Breathing therefore reflects metabolic regulation and coordinated motor control - not a simple readout of how "relaxed" somebody is. [52]

Airway smooth muscle and glands, by contrast, receive autonomic regulation. Vagal parasympathetic cholinergic pathways promote airway narrowing and secretion; inhibitory noncholinergic pathways also exist. Human airway smooth muscle has little or no direct sympathetic innervation, despite possessing receptors through which circulating catecholamines can favor relaxation. A diagram that draws symmetric sympathetic and parasympathetic wires to every human airway therefore misrepresents the anatomy. [11]

Pupils and focusing

The pupil offers a clear example of distinct muscles with different control. Parasympathetic output through the oculomotor nerve and ciliary ganglion contracts the iris sphincter, narrowing the pupil. Sympathetic output activates the dilator muscle. In the light reflex, retinal/optic input is the sensory limb; bilateral midbrain connections support constriction of both pupils. This is not a vagal reflex. [53]

Parasympathetic contraction of the ciliary muscle changes tension on the lens suspension and supports near focusing. Thus "parasympathetic" can mean contraction in one structure while producing a different functional outcome elsewhere. [54]

Gastrointestinal regulation through local and long loops

A particularly informative counterexample to "parasympathetic means contraction" is **gastric accommodation**. As a meal arrives, the proximal stomach relaxes to receive it. Vagal reflex pathways recruit inhibitory enteric neurons, including nitric-oxide signaling. A placebo-controlled human experiment inhibiting nitric-oxide synthesis reduced meal-induced accommodation. Facilitating digestion therefore sometimes requires relaxation, not simply stronger contractions. [55]

The vagus supplies major upper gastrointestinal pathways, but sacral pelvic pathways are important for distal bowel control. Neither "the whole gut is vagal" nor "the enteric system is disconnected from the brain" is an accurate generalization. [56]

Bladder storage and emptying

Bladder function requires coordination among autonomic, sensory and somatic systems. During storage, sympathetic activity favors a relaxed detrusor and increased outlet resistance; pudendal somatic output supports contraction of the external urethral sphincter. During emptying, sacral parasympathetic pelvic pathways contract the detrusor while outlet and external-sphincter activity are appropriately reduced. Stretch-related afferent input informs spinal and supraspinal circuitry, and brainstem/cortical systems help determine when emptying is appropriate. These pelvic parasympathetic commands are not carried by the vagus. [12]

Receptor populations explain different storage effects within the same organ: sympathetic signaling commonly favors detrusor relaxation through beta-3 receptors and outlet contraction through alpha-1 receptors. [57]

The lesson is that a useful action can require simultaneous activation of one muscle and inhibition of another. More activity in all components would not constitute better coordination.

Temperature and sweating

Thermal regulation combines skin/core temperature information with coordinated effector responses. Human eccrine sweating is mainly driven by **sympathetic cholinergic** fibers releasing acetylcholine onto muscarinic receptors. Acetylcholine is therefore not an exclusive marker of parasympathetic action. Sweat production is one output; effective evaporative heat loss additionally depends on environmental conditions. [9]

Skin circulation provides another counterexample to simplistic tables. Human cutaneous sympathetic control includes a noradrenergic vasoconstrictor system and, particularly in nonglabrous skin, an active vasodilator system recruited during heating. The latter involves cholinergic cotransmission and several participating mediators; it is not merely the absence of constriction. Sweating and skin blood flow must also coexist with the need to maintain pressure and perfusion. [13]

Metabolic and reproductive coordination

Adrenal-medullary chromaffin cells receive preganglionic sympathetic input and release epinephrine and norepinephrine into blood. This is a neuroendocrine route: a neural command produces a circulating signal with effects beyond the immediately innervated tissue. It should be distinguished from cortisol production in the adrenal cortex. [58]

Pancreatic islets sense glucose themselves, while central/autonomic inputs modify hormone release. During falling glucose, sympathetic influences can restrain insulin, while both sympathetic and parasympathetic activity can contribute to glucagon secretion. That is functional cooperation rather than obligatory opposition. [59]

Sexual function also illustrates coordinated neural and vascular control. In penile physiology, sympathetic and sacral parasympathetic pathways regulate smooth muscle, while somatic pathways support sensation and pelvic-muscle activity. Nitric-oxide signaling helps relax vascular and erectile-tissue smooth muscle. These processes also interact with hormonal and psychological context rather than depending on one autonomic branch alone. [60]

Immune and neuroendocrine signaling runs both ways

Immune mediators can alter brain function and neuroendocrine output; nervous and endocrine signals, in turn, modify immune activity. The hypothalamic-pituitary-adrenal axis is one endocrine feedback route, distinct from autonomic efferent wiring even though the systems interact. Effects vary with timing, tissue and inflammatory context. [61]

Primary experiments demonstrate real mechanisms without establishing a universal wellness switch. In rats, electrical vagus stimulation reduced aspects of an endotoxin-induced inflammatory response; associated human-cell experiments were conducted in culture, not in patients. [62] A 2024 mouse study identified distinct cytokine-responsive vagal sensory populations and brainstem circuitry that influenced inflammation. These are specific experimental circuits and conditions, not proof that a subjective calm state or a higher cardiac variability measure means the whole immune system is "balanced." [63]

Across these examples, "fight or flight" and "rest and digest" remain introductory mnemonics. The physiological reality is distributed, organ-specific feedback and feedforward control, with responses coordinated over different timescales.

Clinical measurement and evidence limits

Begin with the question not the machine

A neurological investigation asks several different questions: **What changed? Where could the disturbance be? What mechanism could explain it? How urgent is it?** A test can help answer one without settling the others. A scan finding, physiological measurement, symptom description, and disease diagnosis are different objects. Treating them as interchangeable is a category error.

The neurological examination samples cognition, cranial-nerve function, strength, sensation, reflexes, coordination, and gait. Its value is the pattern across findings and their relationship to the history. Magnetic resonance imaging (MRI) provides detailed tissue information using magnetic fields and radiofrequency signals; computed tomography (CT) uses X-rays and can reveal bleeding, structural abnormalities, and injury. Electroencephalography (EEG) records electrical activity through scalp electrodes. These methods are complementary, not competing versions of a single nervous-system health score. MRI findings depend on the anatomical region and sequences examined; neither MRI nor CT directly measures a person's intentions, trustworthiness, or felt safety. [64]

An explicit interpretive rule follows: a normal result means that the particular investigation did not demonstrate its target abnormality under the conditions used. It does not mean that every possible neurological explanation has been excluded. Conversely, an abnormal result is not automatically the cause of the presenting complaint. The missing step is *clinical correspondence*: does the finding explain the location, timing, and character of the problem?

Electrical recordings are not interchangeable

For suspected epilepsy, NICE recommends interpreting EEG with the history and examination. A normal routine EEG does **not** exclude epilepsy; additional recording approaches can be considered when uncertainty remains. Cardiac and metabolic events can resemble seizures, so the diagnostic process also considers electrocardiogram (ECG) findings and metabolic disturbances. MRI investigates possible structural causes rather than serving as an all-purpose seizure detector. [65]

Nerve-conduction studies assess signal speed and amplitude in selected large peripheral motor and sensory nerves. Needle electromyography (EMG) examines muscle electrical activity and helps distinguish patterns arising from nerve or muscle disorders. These are not whole-brain measurements. Conventional large-fiber studies do not directly survey all small sensory and autonomic fibers; investigation may instead involve autonomic tests or skin sampling, depending on the question. Peripheral neuropathy itself can affect motor, sensory, autonomic, or mixed functions. [66]

Hypothetical non-equivalence: an individual could have a normal brain image yet an abnormal peripheral nerve study. That would not be contradictory: the tests concern different structures and properties. Equally, two identical reported pulse rates do not establish identical underlying autonomic activity.

Autonomic testing examines domains not a single balance

A consensus endorsed by the American Autonomic Society, American Academy of Neurology and International Federation of Clinical Neurophysiology separates cardiovascular adrenergic, cardiovagal, and sudomotor domains. Heart-rate responses to breathing and Valsalva contribute cardiovagal information; blood-pressure responses to orthostatic challenge and Valsalva contribute cardiovascular

adrenergic information; sweat testing investigates sudomotor pathways. Interpretation uses reference values, technical quality, and clinical circumstances. A battery helps characterize the distribution and severity of dysfunction because **no single test establishes the entire pattern of autonomic failure**. Orthostatic hypotension concerns an abnormal blood-pressure fall on becoming upright; an autonomic disorder is one possible explanation, not a synonym for every episode of dizziness. [14]

A useful distinction is **response**, **syndrome**, and **cause**. A postural cardiovascular response is something measured; a syndrome requires a defined clinical pattern; a cause explains why that pattern occurs. Naming one does not automatically identify the next. This prevents the inference that any unusual pulse, sweating episode, or gastrointestinal symptom reveals a global failure of one autonomic branch.

Postural orthostatic tachycardia syndrome (POTS) illustrates the distinction. The NIH expert-consensus account requires a sustained heart-rate rise of at least 30 beats/minute within ten minutes upright, or 40 for ages 12-19, alongside orthostatic symptoms, at least three months' duration, absence of orthostatic hypotension, and exclusion of alternative explanations. A single elevated reading is therefore insufficient. Potential mechanisms are heterogeneous; many patients have normal cardiovagal and vasomotor test results, while a subgroup shows evidence associated with small-fiber neuropathy. This is not evidence that their symptoms are imaginary: different tests examine different physiological functions. [67]

Autonomic neuropathy is another example: injury to autonomic peripheral nerves can disturb sweating, vascular regulation, or digestion. It is not the same category as an isolated symptom or a consumer-device label of stress. [66]

Heart rate variability and the scope of vagal tone

Heart-rate variability (HRV) describes variation in successive beat intervals. Respiratory sinus arrhythmia (RSA) is respiratory-linked variation in cardiac timing. Under suitable conditions, RSA-related measures can inform cardiac parasympathetic modulation, but they are not recordings of the entire vagus. Frequency components are affected by physiology and measurement choices; low-frequency power contains mixed influences, so **LF/HF is not a validated universal sympathovagal-balance meter**. Sympathetic and parasympathetic contributions can vary independently, including coactivation and coinhibition. The same cardiac outcome can therefore arise from different combinations of branch activity. [68]

A classic primary study in 29 healthy young adults compared RSA with pharmacologically assessed cardiac parasympathetic control. They were associated but not proportionally interchangeable. Breathing frequency and depth materially influenced the relationship; slow breathing could increase RSA without a uniform change in mean cardiac interval across participants. This supports a limited inference about cardiac regulation, not an unrestricted inference about all vagal output or all people. The study's small, selected population also matters when generalizing. [15]

Analytical implications:

- A metric can be useful without being a direct neural measurement.
- A group-level association does not establish an accurate individual classifier.
- A change in a proxy does not prove a change in every construct associated with it.
- A cardiac measure cannot, by itself, establish personality, moral character, consent, trauma history, interpersonal safety, or the truthfulness of someone's account.

- Calling a measure vagal tone does not expand what it actually samples. The organ, time window, physiological context, and operational definition must remain attached to the claim.

Consider a hypothetical report that an intervention increased HRV and therefore restored a person's capacity to trust. Even if the HRV change were measured correctly, the conclusion would require additional, independently justified evidence about trust. Renaming the physiological variable does not supply that evidence. This is a logical limitation, not a denial that bodies and psychological experiences interact.

Stress physiology is not a moral binary

The sympathetic branch is not inherently harmful, and parasympathetic activity is not inherently beneficial in every circumstance. Adequacy means fitting physiological demand, not maximizing one branch. A simple one-dimensional opposition also misses independent and simultaneous changes in the branches. [68]

Consequently, a framework should not equate arousal with danger, calm appearance with consent, or immobility with a specific brainstem mechanism. Those are interpretations requiring context and separate evidence. Describing someone's experience compassionately does not require presenting an unmeasured neural explanation as established fact.

Contested claims in polyvagal theory

Polyvagal theory connects autonomic organization with defensive behavior and social engagement. A 2026 international expert critique disputes key premises about RSA as an index of central vagal drive, the functions of the nucleus ambiguus and dorsal motor nucleus, vertebrate vagal evolution, and the proposed specificity of mammalian social behavior. These objections concern particular physiological and evolutionary propositions, not the existence or importance of the vagus itself. [17]

Porges's published response argues that the critique misrepresents the theory's articulated claims and disputes its standards of refutation. This establishes an active scientific disagreement, not equal evidential strength for every proposition. A responsible account distinguishes the proponent's position, the critics' specific objections, and independently established autonomic findings. [18]

Two further distinctions are essential. First, evidence that a supportive interaction helps someone would not automatically verify a proposed evolutionary story. Second, a weakness in that evolutionary story would not, by itself, show that every practice marketed using its vocabulary is ineffective. Each intervention needs its own outcome evidence, comparator, population, and harms assessment. Neither blanket endorsement nor blanket dismissal substitutes for claim-by-claim evaluation.

Different categories of central nervous system disease

- **Vascular:** stroke involves interrupted blood supply or bleeding. Sudden focal changes are an emergency, including when they resolve. [69]
- **Episodic electrical disturbance:** suspected seizures require clinical characterization; an event resembling a seizure can have another explanation, and epilepsy assessment is not reducible to one EEG. [65]
- **Neurodegeneration:** progressive loss of neural function can produce cognitive or movement syndromes. Dementia is a syndrome with multiple possible causes; Alzheimer's disease is not a synonym for every cognitive complaint. Vascular and degenerative processes can coexist. [70]

- **Injury:** traumatic injury belongs to a mechanism-based category, while imaging describes detectable tissue consequences. A structural description and a functional examination answer different questions. [64]
- **Infectious/inflammatory:** meningitis affects the coverings of the brain and spinal cord; encephalitis affects brain tissue. Infection is not the only possible inflammatory cause. Rapid progression makes suspected cases urgent. [71]
- **Immune-mediated/demyelinating:** multiple sclerosis involves immune injury to CNS myelin and nerve cells; it is not simply peripheral nerve damage. This labels a mechanism and location, not one symptom pattern. [72]
- **Headache/episodic syndromes:** migraine involves neurological symptoms beyond head pain; some attacks include sensory, visual, or language aura. Migraine is not synonymous with every headache or every transient neurological symptom. [73]

Outside the CNS: myasthenia gravis illustrates impaired transmission at the neuromuscular junction, distinct from motor-nerve injury or primary muscle disease. Autoimmune disruption of signaling to muscle can produce weakness. [74]

This taxonomy is deliberately not a diagnostic checklist. Mechanism, anatomical site, time course, syndrome, and severity are different classification axes. A person can occupy more than one category, and similar symptoms can arise through different mechanisms.

Narrow safety takeaway

Sudden one-sided weakness, new speech or vision difficulty, or abrupt severe headache warrants emergency assessment rather than interpretation through an autonomic-state framework. Symptoms that disappear can still represent an emergency. [69] Fever with severe headache, neck stiffness, altered consciousness, or rapidly worsening neurological symptoms also warrants urgent medical assessment for potentially serious causes. [71]

The general evidence rule is simple: **keep the measurement, the mechanism, the diagnosis, and the human meaning separate until evidence justifies linking them.**

Research interpretation and proposed Blackwish use

What this evidence can support

The factual sections distinguish anatomy, neural activity, physiological output, measured proxies, clinical syndromes and human interpretation. They do not provide a validated classifier of character, consent or trustworthiness.

For Blackwish, the proposed use is to check project text for category errors. Neurological language can describe a mechanism when evidence warrants it; it should not make a metaphor appear experimentally established.

These are proposed interpretive safeguards, not an implemented project revision. No live system, participant experiment or clinical procedure is involved. Explicit manual ratification remains pending for any proposed deployment or project revision.

Textual check one

Claim to test: Parasympathetic activity is always beneficial. Its effects can include slower cardiac pacemaking, pupil constriction or airway narrowing. Value depends on the organ and demand.

Revision to consider: Name the particular response and the context in which it is useful, rather than assigning a global good-versus-bad label.

Textual check two

Claim to test: Increased HRV establishes trust, consent or recovery from trauma. Cardiac timing can change with respiration and measurement conditions without independently observing those constructs.

Revision to consider: Separate the physiological observation from the human conclusion and identify the additional evidence needed. A proxy cannot replace a person's consent.

Textual check three

Claim to test: A new molecular classification or animal mechanism authorizes project-wide revision. Mouse developmental evidence can challenge sacral taxonomy without settling human clinical terminology; an experimental neuroimmune effect does not establish a general wellness intervention.

Revision to consider: Record species, method, result and inference separately. Preserve current text and proposed alternatives until the user ratifies a change. These are bounded textual counterexamples, not experiments or doctrine certification.

Glossary

Definitions here are concise reminders of terms explained in the preceding chapters. They should be read with their anatomical and measurement context.

Term	Meaning in this reference
Acetylcholine	A transmitter used in several central and peripheral pathways, including both autonomic preganglionic divisions.
Afferent	Carrying information toward the central nervous system.
Autonomic	Relating to visceral motor regulation and its associated control networks.
Axon	A neuronal process that carries signals toward other cells.
Baroreflex	A feedback response linking pressure-sensitive afferents with cardiovascular output.
Central nervous system	Brain and spinal cord.
Cholinergic	Using or responding to acetylcholine; not synonymous with parasympathetic.
Circuit	Interconnected cells or regions whose interactions help produce a function.
Efferent	Carrying commands away from the central nervous system.
Enteric nervous system	Local neural and glial networks in the gut wall.
Ganglion	A collection of neuronal cell bodies in the peripheral nervous system.
Glia	Diverse non-neuronal cells contributing to support, immune functions and neural signaling environments.
Heart rate variability	Variation in successive cardiac beat intervals, defined by the recording and metric used.
Homeostatic regulation	Processes that help maintain physiological variables within workable ranges.
Interoception	Sensing, interpreting and regulating the body's internal condition.
Muscarinic receptor	A G-protein-coupled acetylcholine receptor class.
Nerve	A bundle of peripheral axons, often carrying more than one functional kind.
Nicotinic receptor	A ligand-gated acetylcholine receptor class with neuronal and muscle forms.
Norepinephrine	A transmitter commonly used by sympathetic postganglionic neurons, with effects determined by receptors and targets.
Parasympathetic	A conventional visceral motor division with cranial and sacral pathways.
Peripheral nervous system	Nerves and ganglia connecting central structures with the body.
Phasic	Linked to a transient event or burst, rather than maintained baseline activity.
Plasticity	Change in neural properties, without implying unlimited or uniformly beneficial change.
Plexus	An interwoven network of nerves or neural elements.
Postganglionic	Downstream of an autonomic ganglion in the usual efferent pathway.

Term	Meaning in this reference
Preganglionic	Originating centrally and projecting to an autonomic ganglion or its specialized counterpart.
Proprioception	Sensory information concerning body configuration and movement.
Receptor	A molecular structure that responds to a signal and initiates a cellular effect.
Respiratory sinus arrhythmia	Respiratory-linked variation in cardiac timing.
Somatic motor	Output to skeletal muscles, including activity within automatic reflexes.
Sudomotor	Relating to neural control of sweating.
Sympathetic	A visceral motor division with conventional thoracolumbar origins and distributed target pathways.
Synapse	A specialized signaling connection between cells.
Tonic	Maintained baseline activity, assessed in a specified pathway and context.
Vagus	Cranial nerve X, a mixed nerve carrying sensory and motor components, including parasympathetic efferents.

Research roadmap

Start with the scale of the question

For an anatomical question, trace the origin, route, synapse and target. For a physiological question, identify the controlled variable and the feedback that affects it. For a psychological claim, establish how the psychological construct was measured rather than assuming its definition from a physiological proxy. A good explanation can connect these scales without treating them as interchangeable.

Evaluate a paper by its claim and method

Read the actual result before adopting the interpretation. Record whether the work studies humans, animals, isolated tissues or computational models. Distinguish a direct neural recording from an indirect cardiac index. Keep the population, sample size, comparison condition and time window attached to the finding. Ask whether a proposed generalization requires a new measurement or a different population.

Prioritize unresolved questions

Useful next research questions include how organ-specific autonomic outputs are coordinated across states, how afferent pathways shape central regulation, and how measurement methods distinguish mechanisms with similar outward effects. The sacral-classification dispute is a focused example of competing category definitions. HRV interpretation is a focused example of the distance between a valid measurement and an invalid inference.

For any proposed application, separate four records: the empirical finding, the explanatory model, the proposed use, and the authorization to make a change. Evidence may justify one without establishing

the others. This keeps open questions visible and prevents a descriptive research document from silently becoming an intervention or deployment decision.

Evidence and reference notes

This is a structured educational synthesis, not a systematic review, clinical practice guideline or exhaustive catalogue of neurology. Sources include official health agencies, clinical guidelines, investigator reviews, textbooks and selected primary studies. An article's presence in PubMed or PubMed Central does not imply that NIH authored or endorsed its conclusions.

Primary studies are described with their experimental scope where relevant. Reviews organize broader bodies of evidence but do not themselves constitute a new controlled experiment. Critical essays and replies identify competing arguments; their publication does not settle the dispute. The underlying methods and findings deserve more weight than a paper title or a familiar explanatory label.

The source links were checked on 24 October 2026. Some pages were accessible through indexed source text or abstracts rather than complete publisher full text. Bibliographic dates are not invented where living reference pages or textbook metadata did not establish them. No long passages are quoted. The linked sources allow the reader to inspect methods, fuller context and subsequent corrections.

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